ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2022-2023**

DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4107: Structured Prgoramming I

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) In earlier days, applicants had to apply for admission in IUT using SMS. The format for the SMS was IUT <First 3 letters of HSC education board> <HSC Roll> <HSC Passing Year> <First 3 letters of SSC education board> <SSC Roll> <SSC Passing Year>. Each part of the SMS will be separated by a space. A sample SMS is given below: 10 (CO3) (PO2)

   IUT DHA 216754 2018 DHA 206543 2016

   Write a function named *validate*, which receives a string as an argument representing the whole SMS and returns 1, if the SMS is valid, otherwise it returns 0. The SMS will be valid, if the first word is IUT, both SSC and HSC board is DHA (all must be capital letters), and HSC passing year is at-least 2 years more than SSC passing year.

   b) Assume a 2-Dimensional array is used to store the applicant's ID and the total marks obtained in selected subjects (Physics, Chemistry, Mathematics and English). Now answer the following questions: 6 + 9 (CO3) (PO2)

   i. Write a function that receives an integer and a reference to a 2-Dimensional array as an arguments. The integer indicates the total number of applicants and the each row of the array contains information of each candidate(applicant's ID and total marks). The function will sort the array based on the total marks for each applicant.

   ii. Write a function that identifies the list of applicants selected for the admission test. The function receives the sorted array (done by previous function) and selects at-least first 8000 applicants and stores their ID in an array. The first element of that array will be number of applicants selected. The applicants having same marks as 8000th applicant must be selected as well. The function will return the reference to the array used to store the selected applicants.
   For example: consider the following sorted array given in Table 1.

Table 1: Sample Array for Question 1b).

102	99
101	97
105	96
104	96
103	55

Now if you are asked to select top 3 students, in that case you have to select the 4th student as well, as his marks is same as 3rd student's mark. So the content of the resultant array will be: [4, 102, 101, 105, and 104]

2. a) IUT auditorium is one of the several venues for the admission test, which can accommodate 600 applicants. The seats in the auditorium are arranged in 30 rows and 20 columns. Assume you and your friend both got the seats in IUT auditorium, and you want to calculate how far you are from each other. The distance is determined by summing the differences between their respective row and column positions. Write a user-defined function that receives the seat plan and the id of you and your friend as argument and returns the distance. 9 (CO3) (PO2)

   b) The admission test consists of 100 multiple-choice questions (MCQs) with four options each (a, b, c, d). For each correct answer, one mark is awarded, and for each incorrect answer, 0.25 marks are deducted. Unanswered questions receive no penalty. The function you need to write takes two strings as arguments: one representing the applicant's answers (including 'x' for unanswered questions) and the other containing the correct answers. The function calculates and returns the marks obtained by the applicant based on the specified scoring system. 10 (CO3) (PO2)

   c) A unique student ID is assigned to each student when they register for a program. The digits of a student's ID represent various information. For example: first 2 digits indicate the batch number, the fourth and fifth digits combinedly indicate the department (01-MPE, 02-EEE, 03-TVE, 04-CSE, 05-CEE, 06-BTM), the third digit from the last indicates the section etc. Given a student ID as an **integer** input, your task is to print the batch number, department and section of that student. A sample input-output is given in Table 2. 6 (CO3) (PO2)

Table 2: Sample Array for Question 2c).

Input	Output
220042112	Batch: 22, Department: CSE, Section: 1
210021213	Batch: 21, Department: EEE, Section: 2

3. a) Students submit digital pictures renamed with their student IDs (e.g., 220042112.jpg). To create backups, an automated process is needed to generate 20 copies of each image, following to the naming convention "id_backup1.jpg", "id_backup2.jpg", ..."id_backup20.jpg". Write a function that takes a student ID as an argument and makes 20 copies of the picture of that particular student following the naming style mentioned earlier. 14 (CO3) (PO2)

   b) Student's personal information is stored in a CSV file named "info.csv" with fields including Student ID (integer), Student name (string, up to 100 characters), Phone number (string, up to 20 characters), and CGPA (fractional value) separated by a comma. Answer the following questions: 3 + 4 + 4 (CO3) (PO2)

      i. Declare a structure that can store the information of a student and create an array of that structure to store the information of 2000 students.

      ii. Write a function named "load", that reads the information of 2000 students from the "info.csv" file and stores it into the structure array passed as its argument.

      iii. Write a funciton named "save", that overwrites the current content of the "info.csv" file by the information of 2000 students which are passed as an argument to the function.

4. a) Consider the Code Snippet 1, written by a novice programmer as an attempt to print "hello_world" to console. 5 (CO2) (PO1)

```
1  #include <stdio.h>
2  int main()
3  {
4      hello_world("printf()");
5      return 0;
6  }
```

Code Snippet 1: A C program for Question 4.a)

Your task is to add necessary codes in this program without adding, removing, or modifying anything inside the main function, so that the program displays "hello_world" to the console.

b) In C programming, variable names must follow specific rules. Beginners sometimes write invalid variable names. Your task is to create a function that takes a variable name (potentially invalid) as a string argument. The function removes invalid characters without altering the character sequence in given string and prints the largest possible valid variable name. Some sample input-output is given in Table 3. 10 (CO2) (PO1)

Table 3: Sample input-output for Question 4c).

Input	Output
variable#1	variable1
12v@riaBle_2	vriaBle_2

c) Students in a lab task are required to write a program featuring a recursive function that calculates the sum of the digits of an integer. Code Snippet 2 attempts to provide a solution, but the lines of the program are not arranged in the correct order. Rearrange them to produce the desired output. 10 (CO2) (PO1)

```
1  #include<stdio.h>
2  int f1(int num){
3  int f1(int num);
4  int main(){
5  int x, num;
6  int x;
7  scanf("%d",&num);
8  num /= 10;
9  x += f1(num);
10 x = f1(num);
11 x = num%10;
12 x = 0;
13 if(!num)
14 if(num){
15 printf("%d",x);
16 return x;
17 return 0;
18 }
19 }
20 }
```

Code Snippet 2: An unordered C program for Question 4.c)

5. a) The CSE department has chosen to host a picnic and Mr. Khan is responsible for recording first names, while Mr. Ahmed records the last names of the participants. Mr. Khan's keyboard doesn't allow the typing of vowels, and Mr. Ahmed's keyboard is stuck in uppercase. Develop a user-defined function that, when given a participant's first and last name as arguments, displays the stored name, accounting for the specific keyboard limitations. A sample input-output is given in Table 4. 10 (CO2) (PO1)

Table 4: Sample Array for Question 5a).

First Name	Last Name	Output
Kamal	Khan	Kml KHAN
Omi	Azad	m AZAD

b) In response to ongoing strike and road blockade issues, the CSE department considers postponing the picnic. To assist in the decision-making, a function is needed to analyze a given news string for the occurrences of the words "strike" and "blockade." If the combined count of these words exceeds one hundred, the function returns 1, indicating the picnic should be cancelled; otherwise, it returns 0. 10 (CO3) (PO2)

c) The organizer for the picnic will be the person whose name comes first in the lexicographically sorted list of participants. Write a function that takes the list of all participants as an argument and prints the name of the organizer. 5 (CO3) (PO2)

6. a) Students participate in four quizzes, a mid-term exam, and a semester final exam to determine their final grade in a course. The marks for each student are stored using a structure. As new student numbers are changed during the semester, rather than using a fixed-size array, the course teacher has chosen dynamic memory allocation for storing the marks for the students. 5 + 5 (CO3) (PO2)

Now answer the following questions:

i. Write a program that dynamically allocates the memory to store 100 student's marks.

ii. Extend the previously allocated memory, so that 5 additional student's information can be stored.

b) The final grade is calculated by adding the marks of mid-term, final, attendance and the marks of best of 3 quizzes. Consider the simplified grading policy given below: 10 (CO3) (PO2)

- 80-100 percent is: A+
- 70-79 percent is: A
- 60-69 percent is: B
- 50-59 percent is: c
- 40-49 percent is: D
- below 40 percent is: F

Write a function that takes all the marks (including all 4 quiz marks) of a student as an argument and prints the final grade.

c) The course teacher of structured programming has given an encrypted message to all the students. Each character of the message is encrypted by reversing the last bit of the binary representation of each character. The message is given below: 5 (CO3) (PO2)

CDRU!NG!MTBJ!@OE!RNSSX!GNS!UID!DYUS@!BM@RR

Your task is to read the encrypted message and decrypt it to print the original message.